ECI 144 HW 1

Charles Hammond

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Problem 1:

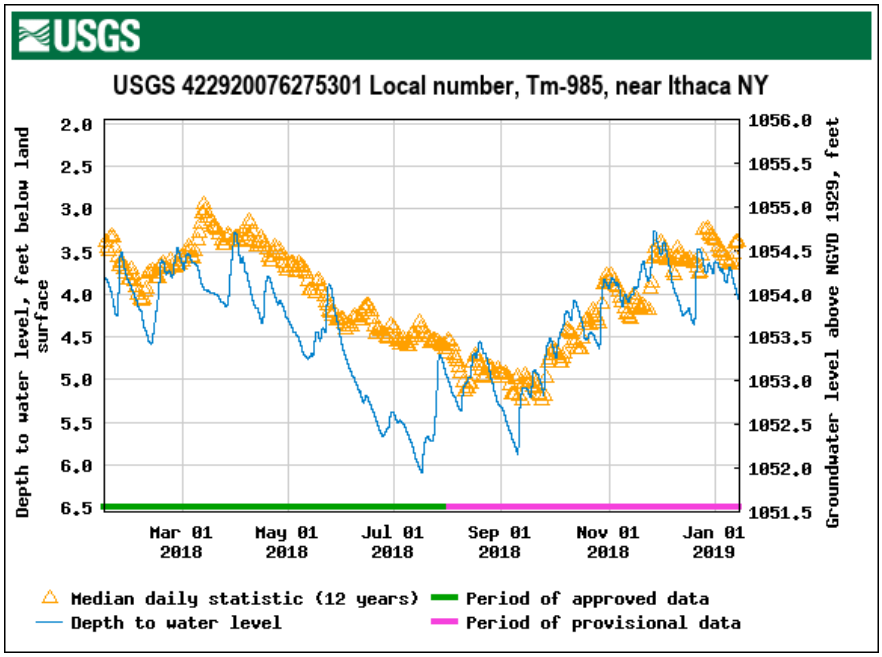

Figure 1: Depth to water values for the last 365 days near Ithaca, NY. Source: USGS National Water Information System

Most recent instantaneous value: 4.07 ft, 01-15-2019 18:15 EST

Minimum: 2.16 ft (2005). Maximum: 4.87 ft (2015).

Average for last 14 years: 3.42 ft (average for only the last year is not an available statistic).

Problem 2:

Diameter=47.5 in, height=10 in.

a) SA=$\frac{\pi}{4}D^2 = 1772.1$ in^2 = 1.14 m^2

b) V=$10in \times SA = 17721.0$ in^3 = 0.3 m^3

c) 11.5 US Gal = 0.435 cubic meters, so: $h = \frac{0.0435m^3}{1.14m^2} = 38.2$ mm

d) In one day the water level dropped by 1.3 gal, so the level in the pan would be $h = \frac{0.039m^3}{1.14m^2} = 34.2$ mm, a drop of 4.0 mm, so the evaporation rate is 4 mm/day.

e) $h = \frac{0.039m^3}{1.14m^2} = 34.2$ mm

f) $h = 34.2mm + 3hr \times 5mm/hr - 1dy \times 4mm/day = 45.2$ mm

g) $45.2mm \times \frac{day}{4mm}$ =11.3 days.

Problem 3:

Using Horton's method and the given parameters, the infiltration capacity curve is

$f_p = f_c + (f_o - f_c)\, e^{-kt} = 17 + 106e^{-0.4t}$ [mm/hr]

For the first storm: the infiltration capacity reaches 50mm/hr at time t=2.92 hrs, which means from t=0 to 2.92 hrs, 50 mm/hr infiltrates, and from t=2.92 to 6 hrs, the infiltration is limited by the infiltration capacity descibed by Horton's equation above.

So: $I = 2.92hr \times 50mm/hr + \int_{2.92}^{6} \left(17 + 106e^{-0.4t}\right) dt = 0.2567m \times 80km^2 \times (1000m/km)^2 = 2.05 \times 10^7 m^3$. The other Storm 1 infiltration values in Table 1 are calculated similarly.

For the second storm: this one shall be done in MATLAB instead of by hand.

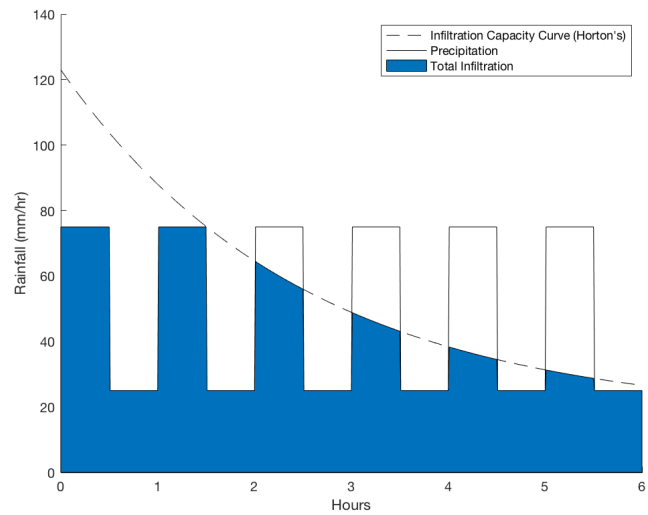

Figure 2: Plot of the infiltration capacity, hyetograph, and total infiltration for the second storm.

Using the trapezoid method of numerical integration, total infiltration is estimated to be 0.2369 m $\times 80km^2 \times (1000m/km)^2 = 1.90 \times 10^7 m^3$ over the 6 hour period. Other Storm 2 infiltration values in Table 1 are calculated similarly.

Table 1: Cumulative Infiltration Values ($\times 10^7 m^3$)

Time (hr)	Storm 1	Storm 2
1	0.40	0.41
2	0.80	0.80
3	1.20	1.14
4	1.54	1.43
5	1.82	1.67
6	2.05	1.90

The results are similar between the two storms because the average hourly rainfall was the same for both storms; however, due to the nature of the capacity curve, the consistent rainfall was able to infiltrate more water than the oscillating rainfall.

Problem 4: Figure 3 shows the given baseflow recession curve with the extrapolated baseflow indicated by a dashed line.

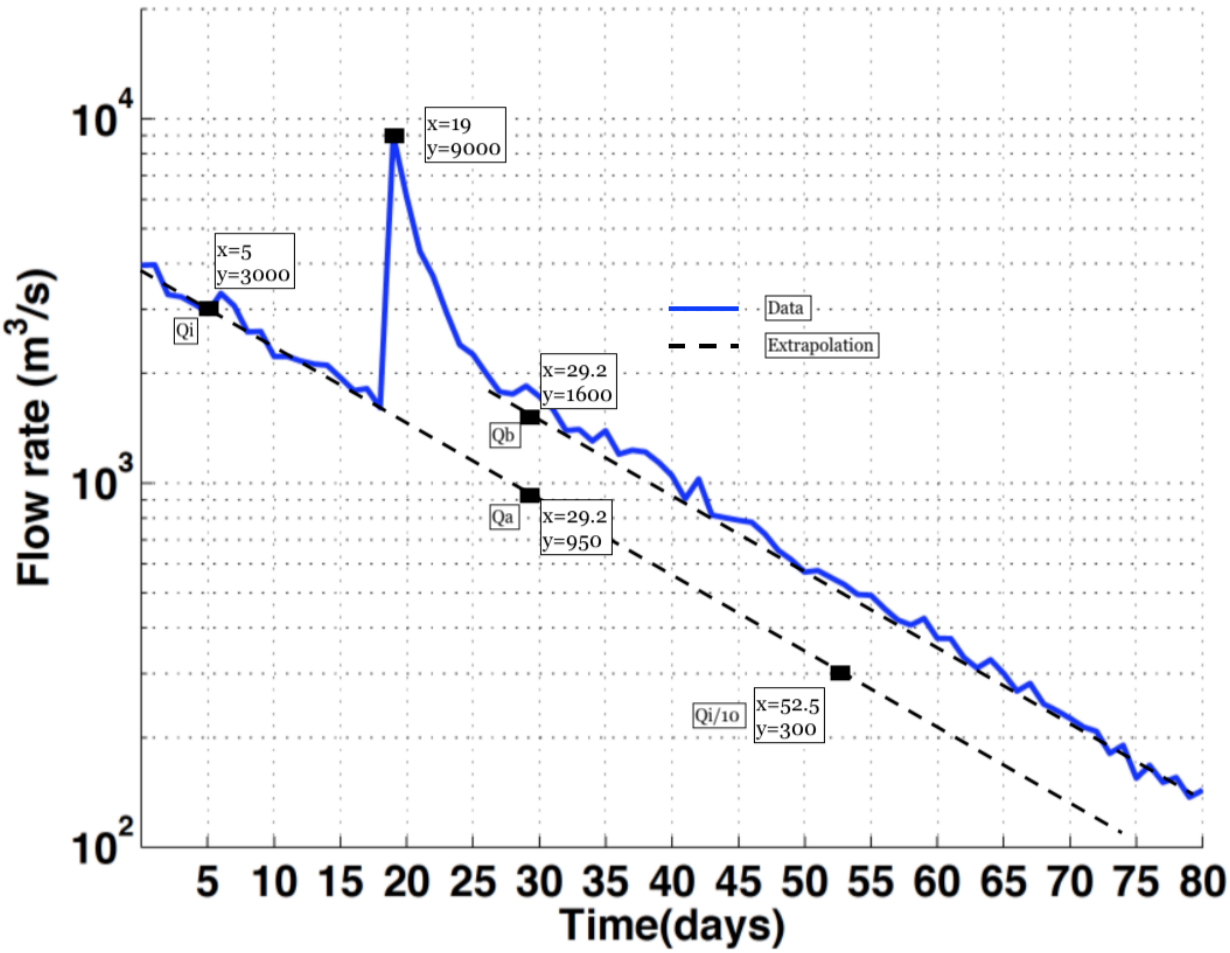

Figure 3: Baseflow recession curve.

$t_1 = 52.5d - 5d = 47.5d$

$t_c = 0.2144t_1 = 10.2d$

So,

$$G = 2\left(\frac{1600m^3/s \times 47.5d}{2.3} - \frac{950m^3/s \times 47.5d}{2.3}\right) \times 24hr/d \times 60min/hr \times 60s/min \rightarrow 2.3 \times 10^9 m^3$$

Problem 5:

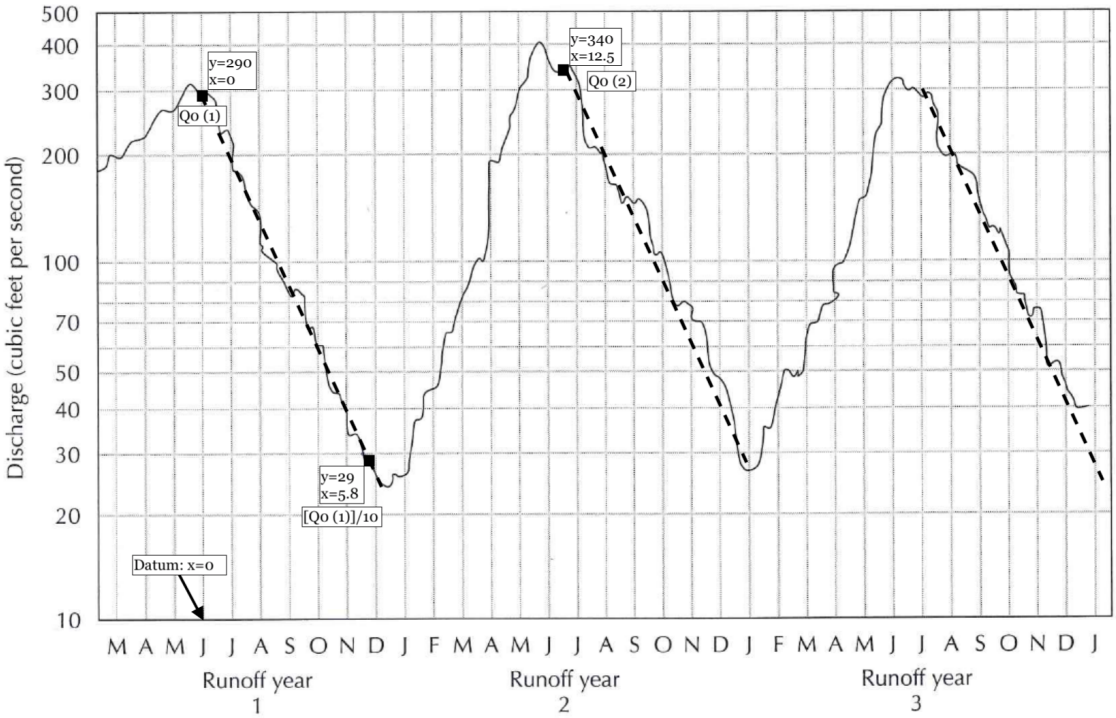

Figure 4: Baseflow recession curve.

So, $t_1 = 5.8$ months (Start of June through October and 80% of November, so that comes out to 30+31+31+30+31+(80% of 30)=177 days)

And $V_{tp}^{(1)} = \frac{290ft^3/s \times 177days}{2.3} \times 24hr/day \times 60min/hr \times 60s/min = 1.9 \times 10^9 ft^3$

And $V_t^{(1)} = \frac{V_{tp}^{(1)}}{10^{t/t_1}} = \frac{1.9 \times 10^9 ft^3}{10^{186.1days/177days}} = 1.7 \times 10^8 ft^3$

Where t=duration of the recession (1), which lasts 6.1 months, which comes out to 30+31+31+30+31+30+10% of 31 = 186.1 days.

So, Recharge (year 1 to 2) =
$V_{tp}^{(2)} - V_t^{(1)} = \frac{340ft^3/s \times 177days \times 24hr/day \times 60min/hr \times 60s/min}{2.3} - 1.7 \times 10^8 ft^3 = 2.1 \times 10^9 ft^3$

Problem 6:

Given the parameters listed:

$A_1 = 1m^2$ and $A_2 = 1.5m^2$

$S_1 = 10m/500m = 0.02m/m$

$n_1 = 0.025$ (from Chapter 1 slides)

a) Using Manning's Equation, $V_1 = \frac{R_1^{2/3} S_1^{1/2}}{n_1} = \frac{\left(\frac{1m^2}{3m}\right)^{2/3} (0.02)^{1/2}}{0.025} = 2.72m/s$

b) $Q_1 = A_1 \times V_1 = 2.72m^3/s$

c) $V_2 = Q_1/A_2 = 1.81m/s$ Since the flow rates are the same, just divide the flow rate by the new area to get the velocity.

d) $S_2 = \left(\frac{V_2 n}{R_2^{2/3}}\right)^2 = \left(\frac{1.81m/s \times 0.025}{\left(\frac{1.5m^2}{3.7m}\right)^{2/3}}\right)^2 = 0.00682m/m$

e) This would increase Manning's n to 0.035 and therefore decrease the flow velocity.

f) $V_1 = \frac{R_1^{2/3} S_1^{1/2}}{n_1} = \frac{\left(\frac{1m^2}{3m}\right)^{2/3} (0.02)^{1/2}}{0.035} = 1.94m/s$

Therefore, $Q_1 = A_1 \times V_1 = 1.94m^3/s$

and $V_2 = Q_1/A_2 = 1.3m/s$, a change of -0.64 m/s.